Master's Comprehensive Examination

Part I | Theory (90 minutes) | March 6, 1993

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a uniform $(\theta, 0)$ distribution with an unknown $\theta < 0$.

(a) Find the MLE of θ.

(b) Show that the MLE is biased.

2. Consider a two-by-two classification of individuals in a population with probabilities

π_{11}	π_{12}
π_{21}	π_{22}

Suppose a random sample of size n was taken from the population and the following contingency table was obtained

n_{11}	n_{12}
n_{21}	n_{22}

(with $n_{11} + n_{12} + n_{21} + n_{22} = n$).

(a) Find the joint distribution of $(n_{11}, n_{12}, n_{21}, n_{22})$.

(b) Find the MLE of the unknown parameter θ under the hypothesis H_0: $\pi_{11} = \theta^2$, $\pi_{12} = \pi_{21} = \theta(1-\theta)$, and $\pi_{22} = (1-\theta)^2$.

(c) Derive the likelihood-ratio-test statistic λ for H_0, and find the number of degrees of freedom for $-2\log(\lambda)$.

3. There are 15 male and 5 female students in Class 1, 5 male and 5 female students in Class 2, and 20 male and 20 female students in Class 3. One person is selected at random from each class to form a committee, and then a chair is selected at random from this committee of 3.

(a) What is the probability that the chair is a male student?

(b) Conditionally on the fact that the chair is a female student, what is the probability that she is from Class 1?

4. Let $X_1, X_2, \ldots, X_n$ be a random sample from the density

$$f(x|\theta) = \theta e^{-x\theta}, \; x > 0, \; \theta > 0.$$

Define $S = X_1 + X_2 + \cdots + X_n$.

(a) Find the conditional distribution of X_1 given $S = s$ for the case $n = 2$.

(b) Let $Y_i = X_i/S$, $1 \le i \le n$. Is S statistically independent of $Y_1, Y_2, \ldots, Y_n$? Why?

(c) Is S sufficient? Why?

5. Consider the linear regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \ 1 \le i \le 2,$$

where x_i are known constants and ϵ_i are independent normal random variables with $E\epsilon_i = 0$ and a common variance $E\epsilon_i^2 = \sigma^2 > 0$. Let $\hat{a}$ and $\hat{b}$ be the least squares estimates of α and β respectively.

(a) Find the variance of $\hat{a}$.

(b) Find the covariance of $\hat{a}$ and $\hat{b}$.

(c) Find a condition on (x_1, x_2) under which $\hat{a}$ and $\hat{b}$ are independent. Explain why?

Master's Comprehensive Examination

Part II — Applied (90 minutes) — March 6, 1993

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. A simple linear regression model

$$Y = \alpha + \beta X + \epsilon$$

was fit to a set of data

$$n = 13 \qquad \sum XY = 514.80$$
$$\sum X = 130 \qquad \sum X^2 = 1562$$
$$\sum Y = 44.4 \qquad \sum Y^2 = 171.30$$

a. Find the estimated regression line.

b. If it is assumed that ϵ is distributed as $N(0, \sigma^2)$, then estimate σ^2.

c. Test the hypothesis that $\beta = 0$. Let the probability of a type I error $= .01$.

d. Find a 95% confidence interval for the mean value of Y when $X = 14$.

e. Find the coefficient of determination. What does this mean?

f. Find an estimate of the correlation coefficient, ρ.

2. A four factor factorial experiment has been run. The factors and the number of corresponding levels are shown below:

A	2 levels
B	3 levels
C	4 levels
D	5 levels

Using our usual notation this means that the experiment was a $2 \times 3 \times 4 \times 5$ factorial. For each treament combination two observations were collected

1. How many treatment combinations does the experiment have?

2. How many data points will be collected for the experiment?

3. Write out the analysis of variance table including the sources of variation, the degrees of freedom, and the expected mean squares.

3. To investigate the relative distribution of synapses on dendritic shafts and spines, samples were taken from ventromedial nuclei of male and female rats. The data based on five male rats and five female rats, are given below:

Ratio of the Number of Nonamygdalid Synapses on Dendritic Shafts to the Number on Dendritic Spines for the Ventromedial Nucleus

Female	Male
6.6	6.1
5.1	3.9
4.5	3.7
4.0	3.4
3.8	4.7

(a) Use a non-parametric test to test for differences between population means.

(b) Give a 95% non-parametric confidence interval for the differences between population means.

4. In a large legislative body, 50% of the members are age 35 to 55, 25% are age 56 to 64, and 25% are age 65 or older. In a sample of 20 people chosen to form a key committee, 5 people are age 35-55, 8 people are age 56-64, and 7 people are age 65 or older. It is claimed that the 20 people were chosen completely at random from the legislative body. Carry out a test of this claim.

5. Let P denote the probability that a spun penny will come up tails. We seek to test the Null Hypothesis: $P = 0.5$, versus the Alternative Hypothesis $p > 0.5$. We are going to spin the penny 100 times and observe X, the total number of tails in the 100 spins. The test will reject the Null Hypothesis if X exceeds some critical number C.

(a) Find C such that the test will have a 5% level of significance.

(b) For the value of C in part (a), find the Power of the test against the alternative $P = 0.60$.

(c) If $X = 60$, what would be the observed level of significance (p-value)?